

Power of Point, Linearity and Coaxiality

Dorian Croitoru

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1 Preliminaries

Recall that if we have a line l given in coordinates by the equation $ax + by + c = 0$ with $a^2 + b^2 = 1$, then for any point $P(x_0, y_0)$, $ax_0 + by_0 + c$ represents the signed distance from P to l . This behaves linearly in $\vec{P}$ in the sense that

$$d_l(t\vec{P}_1 + (1-t)\vec{P}_2) = td_l(\vec{P}_1) + (1-t)d_l(\vec{P}_2).$$

Similarly, the equation of the circle ω with center $O(x_0, y_0)$ and radius r is

$$f(x, y) := (x - x_0)^2 + (y - y_0)^2 - r^2 = 0.$$

For any point $P(x, y)$ one sees that $f(P) = OP^2 - r^2 =: \mathbb{P}_\omega(P)$ is the power of the point P with respect to ω . It is a sort of measure of the position of P with respect to ω .

Given two non-concentric circles ω_1, ω_2 with equations

$$f_i(x, y) = x^2 + y^2 + a_i x + b_i y + c_i = 0, \quad i = 1, 2 \quad (1)$$

any linear combination $\alpha f_1(x, y) + \beta f_2(x, y) = 0$ is the equation of either line (for $\alpha + \beta = 0$) – the radical axis of ω_1 and ω_2 , or (for $\alpha + \beta = 1$) another circle coaxial with ω_1, ω_2 . The family $(\omega_\alpha)_{\alpha \in \mathbb{R}}$ of circles with equations $\alpha f_1(x, y) + (1 - \alpha)f_2(x, y) = 0$ is the pencil of circles generated by ω_1, ω_2 . This includes circles of zero or infinite radius. We list several properties:

1. The radical axis d of ω_1, ω_2 is also the radical axis of any 2 circles of the pencil. In particular, for any $P \in d$, the power of a point $\mathbb{P}(P, \omega_\alpha)$ does not depend on α .
2. Any 2 distinct circles from the pencil generate the same pencil.
3. The centers of all circles in a pencil are collinear on a line perpendicular to their common radical axis.
4. If a point M lies on any two circles of the pencil, it lies all of them.
5. For any point $M \notin \omega_1 \cap \omega_2$, exactly one circle in the pencil passes through M .

This behaves linearly as a function of P . While the $\mathbb{P}(X, \omega)$ is not linear directly in X , we have:

Theorem 1 (Linearity of Power of a Point). *The function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by $F(X) = \mathbb{P}(X, \omega_1, \omega_2) := \mathbb{P}(X, \omega_1) - \mathbb{P}(X, \omega_2)$ is linear.*

Proof. Keeping the notation from (1), for a point $P(x, y)$ we have

$$F(P) = \mathbb{P}(P, \omega_1) - \mathbb{P}(P, \omega_2) = (a_1 - a_2)x + (b_1 - b_2)y + (c_1 - c_2)$$

which satisfies $F(kA + (1 - k)B) = kF(A) + (1 - k)F(B)$ for any $A, B \in \mathbb{R}^2, k \in \mathbb{R}$. □

Theorem 2 (Forgotten Coaxiality Lemma). *Given a parameter $\lambda \neq 0$, the set of points X such that*

$$\mathbb{P}(X, \omega_1) = \lambda \mathbb{P}(X, \omega_2)$$

is a circle coaxial with ω_1, ω_2 . The limiting case $\lambda = 1$ corresponds to the radical axis of ω_1, ω_2 .

We will refer to the circle in the theorem as the *Apollonius λ -circle* of (ω_1, ω_2) .

Corollary 1 (Coaxiality Lemma). *Suppose $\omega_1, \omega_2, \omega_3$ are circles and there exist two points $A, B \in \omega_3$ such that*

$$\frac{\mathbb{P}_{\omega_1}(A)}{\mathbb{P}_{\omega_2}(A)} = \frac{\mathbb{P}_{\omega_1}(B)}{\mathbb{P}_{\omega_2}(B)}$$

Then $\omega_1, \omega_2, \omega_3$ are coaxial.

2 Worked Examples

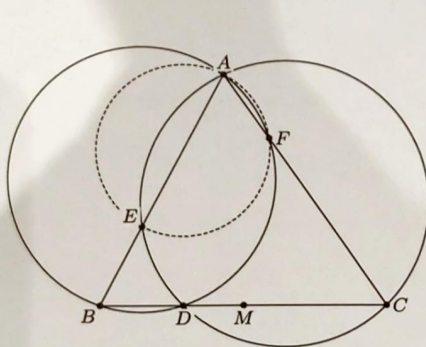
Example 2 (ELMO Shortlist 2013/G3). In $\triangle ABC$, a point D lies on line BC . The circumcircle of ABD meets AC at F (other than A), and the circumcircle of ADC meets AB at E (other than A). Prove that as D varies, the circumcircle of AEF always passes through a fixed point other than A , and that this point lies on the median from A to BC .

Solution. Let M denote the midpoint of BC and $\omega = (ABC)$. We wish to show that $\omega_1 = (AEF)$ always passes through some fixed point P on AM . The idea is to show that $\mathbb{P}(M, (AEF))$ is fixed. Since $\vec{M} = \frac{1}{2}\vec{B} + \frac{1}{2}\vec{C}$, linearity of power of a point gives us:

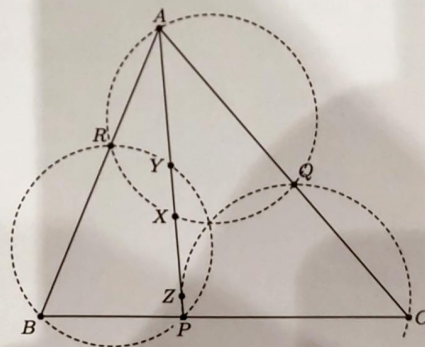
$$\mathbb{P}(M, \omega_1, \omega) = \frac{\mathbb{P}(B, \omega_1, \omega) + \mathbb{P}(C, \omega_1, \omega)}{2} = \frac{BE \cdot BA + CF \cdot CA}{2} = \frac{BD \cdot BC + CD \cdot BC}{2} = \frac{BC^2}{2}.$$

$$\implies \mathbb{P}_{\omega_1}(M) - \frac{BC^2}{4} = \mathbb{P}(M, \omega_1, \omega) = \frac{BC^2}{2} \implies MP \cdot MA = \mathbb{P}_{\omega_1}(M) = \frac{3BC^2}{4} = \text{constant}.$$

Therefore, the point $P = AM \cap \omega_1$ is fixed. □



(a) ELMO Shortlist 2013



(b) USAMO 2013

Example 3 (USAMO 2013/1). In triangle ABC , points P, Q, R lie on sides BC, CA, AB respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ , respectively. Given that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z , respectively, prove that $YX/XZ = BP/PC$.

Proof. Let $BP/PC = \lambda$. Then we let A denote the circle of radius 0.

$$\begin{aligned} PA \cdot AX &= \mathbb{P}(P, A, \omega_A) = \lambda \mathbb{P}(C, A, \omega_A) + (1 - \lambda) \mathbb{P}(B, A, \omega_A) \\ &= \lambda(CQ \cdot CQ - CA^2) + \lambda(BR \cdot BQ - BQ^2) \\ &= \lambda(CA \cdot AQ) + (1 - \lambda)(BA \cdot AR) \\ &= \lambda(CA \cdot AQ - BA \cdot AR) + BA \cdot AR \end{aligned}$$

Thus,

$$\lambda = \frac{PA \cdot AX - BA \cdot AR}{CA \cdot AQ - BA \cdot AR} = \frac{PA \cdot AX - AY \cdot AP}{AX \cdot AP - AY \cdot AP} = \frac{AX - AY}{AX - AY} = \frac{XY}{YZ}$$

as desired. $\square$

Example 4 (USAMO 2015/2). Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at $S \neq A$. Point T lies on arc AQB of ω such that $XT \perp AX$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

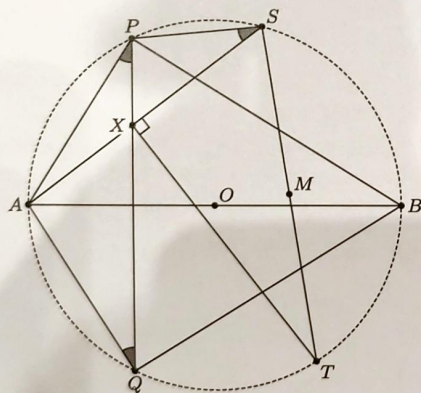
Proof. Let γ be the circle of diameter AO . We will prove that $\mathbb{P}(M, \gamma)$ is fixed. Since $\vec{M} = \frac{1}{2}\vec{S} + \frac{1}{2}\vec{T}$, we have

$$\mathbb{P}_\gamma(M) - MS \cdot MT = \mathbb{P}(M, \gamma, \omega) = \frac{1}{2}\mathbb{P}(S, \gamma, \omega) + \frac{1}{2}\mathbb{P}(T, \gamma, \omega) \implies$$

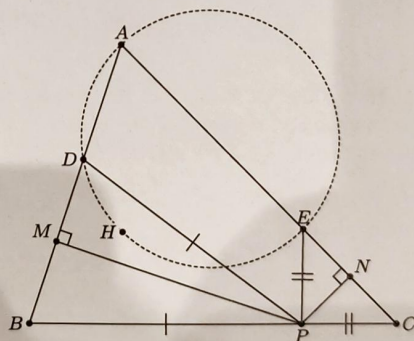
$$\mathbb{P}_\gamma(M) = \frac{1}{4}(AS^2 + AT^2 - ST^2) = \frac{1}{2}AS \cdot AT \cos \angle SAT = \frac{AS \cdot AX}{2}.$$

But $\angle APX = \angle AQP = \angle ASP \implies AX \cdot AS = AP^2$, which is fixed.

Therefore, M lies on a fixed circle centered at the midpoint of $\overline{AO}$. $\square$



(a) USAMO 2015



(b) USA December TST 2012

Example 5 (USA December TST 2012). In acute triangle ABC , $\angle A < \angle B$ and $\angle A < \angle C$. Let P be a variable point on side BC . Points D and E lie on sides AB and AC , respectively, such that $BP = PD$ and $CP = PE$. Prove that as P moves along side BC , the circumcircle of triangle ADE passes through a fixed point other than A .

Solution. Let M, N be the midpoints of BD and CE . Let Γ and ω be the circumcircles of $\triangle ABC$ and $\triangle ADE$. We prove ω passes through H , the orthocenter of $\triangle ABC$. Recall that

$$\vec{H} = \frac{1}{\tan B \tan C} \vec{A} + \frac{1}{\tan C \tan A} \vec{B} + \frac{1}{\tan A \tan B} \vec{C}$$

i.e. H has barycentric coordinates $\left(\frac{1}{\tan B \tan C}, \frac{1}{\tan C \tan A}, \frac{1}{\tan A \tan B}\right)$. By linearity of power of the point,

$$\mathbb{P}(H, \omega, \Gamma) = \frac{\tan A \cdot \mathbb{P}(A, \omega, \Gamma) + \tan B \cdot \mathbb{P}(B, \omega, \Gamma) + \tan C \cdot \mathbb{P}(C, \omega, \Gamma)}{\tan A \tan B \tan C} \implies$$

$$\tan A \tan B \tan C \cdot \mathbb{P}(H, \omega, \Gamma) = \tan B(BD \cdot BA) + \tan C(CE \cdot CA) = 4[ABC].$$

This is fixed, implying $\mathbb{P}(H, \omega)$ is constant. When P is the midpoint of BC , ω is the circle of diameter AH , so $\mathbb{P}(H, \omega) = 0$. Thus, $H \in \omega$ for any P . $\square$

Example 6 (IMO Shortlist 2012/G6). Let ABC be a triangle with circumcenter O and incenter I . The points D, E, F on sides BC, CA, AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of $\triangle BFD$ and $\triangle CDE$ intersect at $P \neq D$. Prove that $OP = OI$.

Solution. Let $\omega, \omega_A, \omega_B, \omega_C$ denote the circumcircles of $\triangle ABC, \triangle AEF, \triangle BFD, \triangle CDE$. Suppose P has barycentric coordinates (x, y, z) with respect to $\triangle ABC$. Applying the linearity property (note that $A \in \omega \cap \omega_A$ can be ignored), we have

$$-\mathbb{P}_\omega(P) = \mathbb{P}_{\omega_A}(P) - \mathbb{P}_\omega(P) = y(BF \cdot BA) + z(CE \cdot CA) \implies$$

$$-a\mathbb{P}_\omega(P) = \mathbb{P}_\omega(P) - \mathbb{P}_\omega(P) = yac \cdot BF + zab \cdot CE$$

Summing the last relation cyclically,

$$-\mathbb{P}_\omega(P) \sum_{\text{cyc}} a = abc(x + y + z) = abc \implies \mathbb{P}_\omega(P) = \frac{-abc}{a + b + c} = OI^2 - R^2 = \mathbb{P}_\omega(I)$$

which implies $OP = OI$. $\square$

Exercise 1 (IMO Shortlist 2012). Let ABC be a triangle with circumcircle ω and l be a line not intersecting ω . Let P be the projection onto l of the center of ω . Let X, Y, Z be the intersections of BC, CA, AB with l . Prove that the circles $(AXP), (BYP), (CZP)$ either have a second common point or are tangent at P .

Hint. Let $D = AB \cap (BYP) \neq B$ and $E = AC \cap (CZP) \neq C$. By the Forgotten Coaxiality Lemma, it's enough to show that

$$\frac{\mathbb{P}_{(BYP)}(A)}{\mathbb{P}_{(CZP)}(A)} = \frac{\mathbb{P}_{(BYP)}(X)}{\mathbb{P}_{(CZP)}(X)} \iff \frac{AD \cdot AB}{AC \cdot AE} = \frac{XP \cdot XY}{XP \cdot XZ}$$

By Menelaus in $\triangle AYZ$ and transversal $B - C - X$ we have $\frac{XY}{XZ} = \frac{AB \cdot CY}{BZ \cdot AC}$, so it suffices to show that $AD \cdot BZ = AE \cdot CY$. $\square$

3 Practice Problems

Power of Point

Problem 1 (Polish MO 2018). In acute $\triangle ABC$, $AB < AC$. Points E and F are feet of altitudes from B and C . The tangent at A to $C(ABC)$ intersects BC at P . The parallel line through A to BC intersects EF at Q . Prove that PQ is perpendicular to the median from A in $\triangle ABC$.

Problem 2 (CGMO 2015). Let Γ_1, Γ_2 be non-overlapping circles and AB and CD be external and respectively, internal common tangents such that $A, C \in \Gamma_1$ and $B, D \in \Gamma_2$. Let $E = AC \cap BD$ and $F \in \Gamma_1$. The tangent through F to Γ_1 intersects the perpendicular bisector of EF at M . Show that MF equals the length of the tangent from M to Γ_2 .

Problem 3 (Iran TST 2011). In acute triangle ABC , $\angle B > \angle C$, M is the midpoint of BC and E, F are the feet of the altitudes from B, C , respectively. Let K, L be the midpoints of ME, MF , respectively and let $T \in KL$ be such that $TA \parallel BC$. Prove that $TA = TM$.

Problem 4. In triangle ABC , points E and F lie on sides AC and BC such that segments AE and BF have equal length. Let (ACF) and (BCE) intersect for a second time at D . Prove that CD bisects $\angle ACB$.

Problem 5 (USAJMO 2012). Given triangle ABC , let P and Q be points on segments $\overline{AB}$ and $\overline{AC}$ such that $AP = AQ$. Let S and R be distinct points on segment $\overline{BC}$ such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R, S are concyclic.

Problem 6 (2016 Taiwan TST). Let O be the circumcenter of $\triangle ABC$, and ω be the circumcircle of $\triangle BOC$. Line AO intersects ω again at G . Let M be the midpoint of BC , and the perpendicular bisector of BC meets ω at O and N . Prove that the midpoint of segment AN lies on the radical axis of the circumcircle of $\triangle OMC$ and the circle with diameter AO .

Problem 7. In $\triangle ABC$, O is the circumcenter, M the midpoint of BC and H is the projection of A onto BC . Line OH meets AM at P . Prove that P lies on the radical axis of (BOC) and the nine-point circle of $\triangle ABC$.

Problem 8 (USA TSTST 2015). Let $\triangle ABC$ be scalene. Let K_a, L_a, M_a be the intersections with BC of the internal angle bisector, external angle bisector, and median from A . The circumcircle of AK_aL_a intersects AM_a again at X_a . Define X_b, X_c analogously. Prove that the circumcenter of $\triangle X_aX_bX_c$ lies on the Euler line of $\triangle ABC$.

Problem 9. Let Γ and ω be two circles tangent at A . Let B, C be on Γ with tangents BY, CZ to ω . Prove that $\frac{AB}{AC} = \frac{BY}{CZ}$.

Linearity of Power of Point

Problem 10. In $\triangle ABC$, let M be the midpoint of BC , D the foot of the altitude from A and O the circumcenter. Prove that $P = AM \cap OD$ lies on the radical axis of (BOC) and the Euler circle of $\triangle ABC$.

Problem 11 (IMO Shortlist 2004/G7). For $\triangle ABC$, let X be on line BC with C between B and X . The incircles of $\triangle ABX$ and $\triangle ACX$ intersect at P and Q . Prove that line PQ passes through a fixed point.

Problem 12 (RMM Shortlist 2017). Let $ABCD$ be a convex quadrilateral and let P and Q be variable points inside this quadrilateral such that $\angle APB = \angle CPD = \angle AQB = \angle CQD$. Prove that the lines PQ obtained in this way all pass through a fixed point or are all parallel.

Problem 13. In trapezoid $ABCD$ with bases AB and CD , points P and Q are chosen such that $\angle BPC = \angle BQC = 180^\circ - \angle DPA = 180^\circ - \angle DQA$. If $U = AC \cap BD$ and $V = BC \cap DA$, prove that PQ passes through the projection of U onto the line through V and parallel to AB .

Problem 14 (Ukraine TST 2013). Let A, B, C, D, E, F be six points, no three collinear and no four concyclic. Let P, Q, R be the intersection points of the perpendicular bisectors of pairs of segments $(AD, BE), (BE, CF), (CF, DA)$, and P', Q', R' be the intersection points of the perpendicular bisectors of pairs of segments $(AE, BD), (BF, CE), (CA, DF)$. Show that $P \neq P', Q \neq Q', R \neq R'$ and prove that PP', QQ', RR' are concurrent or all parallel.

Problem 15 (IMO 2019). Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA , and AB at D, E , and F , respectively. The line through D perpendicular to EF meets ω at R . Line AR meets ω again at P . The circumcircles of triangles PCE and PBF meet again at Q . Prove that lines DI and PQ meet on the line through A perpendicular to AI .

Coaxiality

Problem 16. Beetles A, B move with constant speeds along two lines intersecting at S . Prove that all circles (ASB) are coaxial.

Problem 17. Let AD, BE, CF be the altitudes of a scalene triangle ABC with circumcenter O . Prove that the circles $(AOD), (BOE)$ and (COF) have a second common point $X \neq O$.

Problem 18 (Ceva's theorem on coaxial circles). Given circles w_1, w_2, w_3 and $\lambda_1, \lambda_2, \lambda_3 \neq 0$, let γ_1 be the λ_1 -Apollonius circle of (w_2, w_3) , and define γ_2, γ_3 similarly. Then $\gamma_1, \gamma_2, \gamma_3$ are coaxial if and only if $\lambda_1 \lambda_2 \lambda_3 = 1$.

Problem 19. Given a triangle ABC with a transversal $A' - B' - C'$ ($A' \in BC, B' \in AC, C' \in AB$).

1. The circles with diameters $[AA'], [BB'], [CC']$ are coaxial. In particular, the midpoints of AA', BB', CC' are collinear (Newton-Gauss)
2. The common radical axis of the above circles passes from the orthocenters of triangles $ABC, AB'C', BC'A', CA'B'$.

Problem 20 (Key Poncelet Lemma). Suppose circle ω is inside circle Ω and suppose A_0, A_1, B_0, B_1 are points such that A_0A_1 and B_0B_1 are tangent to ω at X_1, Y_1 , respectively. If X_1Y_1 intersects A_0B_0 and A_1B_1 at P_0 and P_1 respectively, prove that there exists a circle γ coaxial with Ω and ω which touches A_0B_0 at P_0 and A_1B_1 at P_1 .

Problem 21 (Poncelet's Theorem). Let two circles be given on the plane, with the second circle lying inside the first. Suppose there exists a closed n -link polygonal chain (trajectory) whose vertices lie on the first circle and all edges are tangent to the second circle. Then there are infinitely many such closed n -link polygonal chains, and any point on the first circle can be a vertex of such a chain.

Problem 22. Let $ABCD$ be a quadrilateral inscribed in circle ω , and $E = AC \cap BD$. Let $T \in \omega$ be a point of the mixtilinear circle tangent to ω , $[AE]$ and $[BE]$. Prove that there is a mixtilinear circle tangent to ω at T as well as the $[AD]$ and $[BC]$.

Problem 23 (IMO Shortlist 2017). A convex quadrilateral $ABCD$ has an inscribed circle with center I . Let I_a, I_b, I_c, I_d be the incenters of triangles DAB, ABC, BCD and CDA respectively. Suppose that the common external tangents of (AI_bI_d) and (CI_bI_d) met at X , and the common external tangents of (BI_aI_c) and (DI_aI_c) meet at Y . Prove that $\angle XIY = 90^\circ$.